



Universidade do Minho
Escola de Engenharia

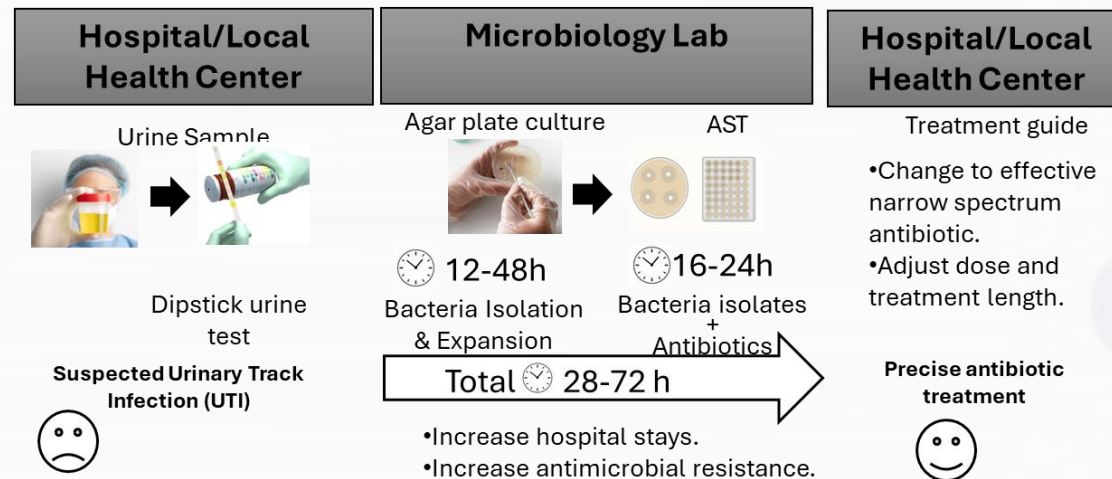
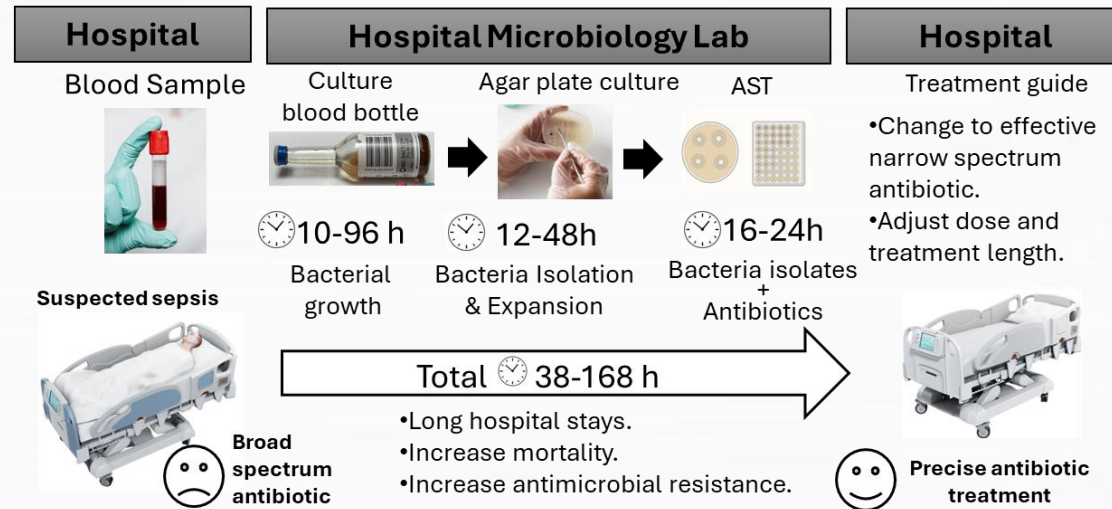


Machine Learning approaches for accelerating antibiotic susceptibility testing (AST) in microfluidic systems

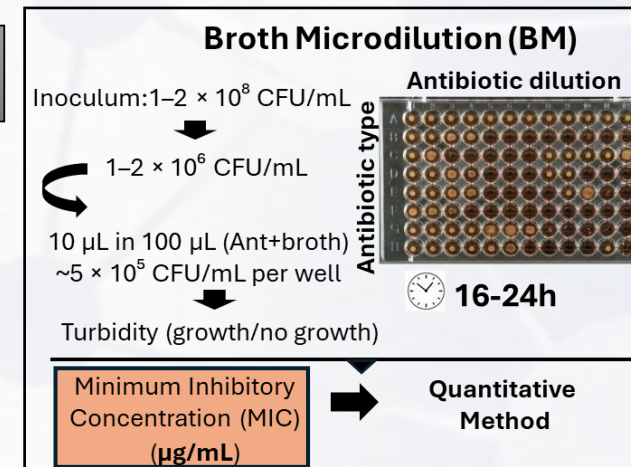
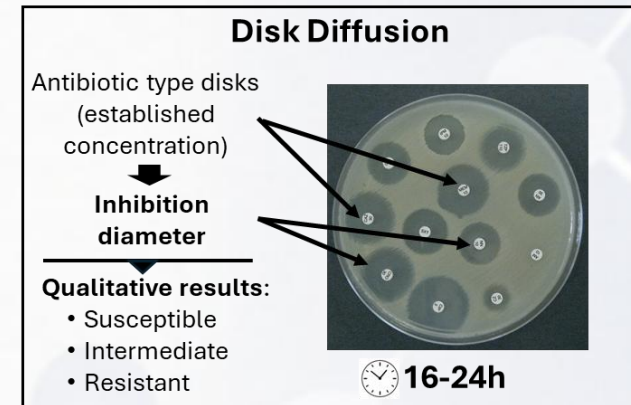
Eduarda Santos

PG 58815

Why is AST important?

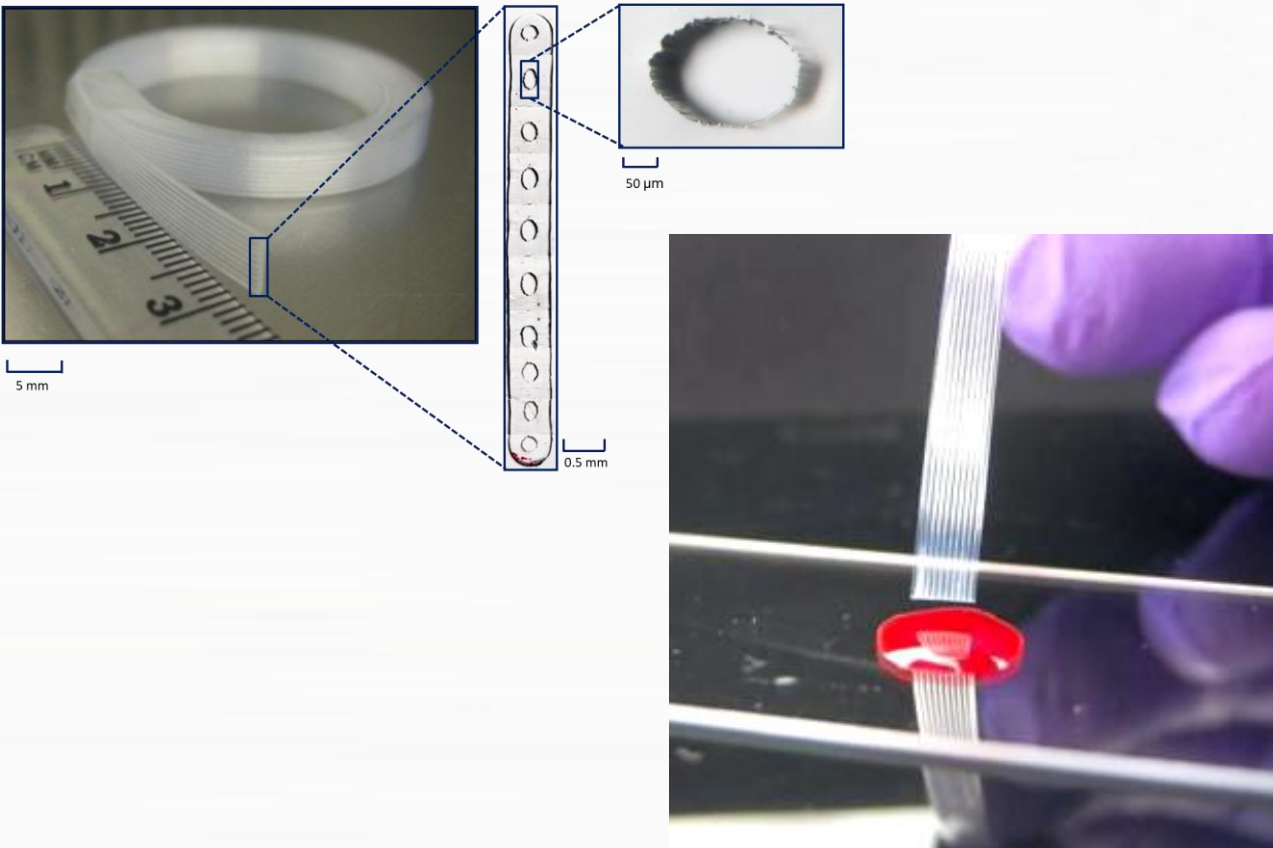


Phenotypic Antibiotic Susceptibility Testing (AST)



MIC value

New technologies for AST



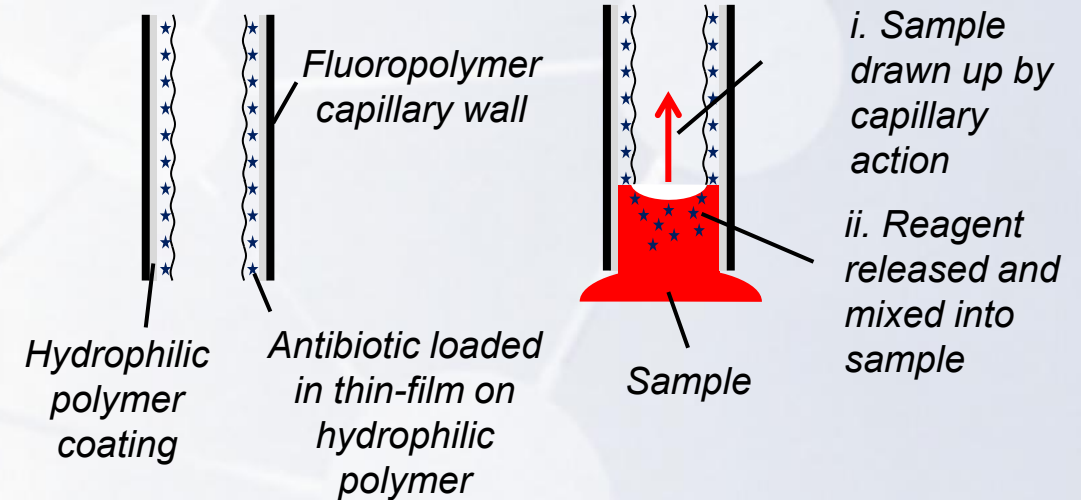
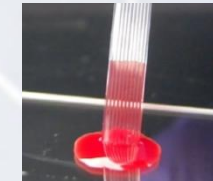
Reis *et al.*, Lab Chip, 2016

Continuous MicroCapillary Film (MCF)

Step 1: Bacteria loaded test dip strip



Step 2: Test strip dipped into sample



Rapid AST in the Microcapillary Film

$\sim 5 \times 10^5$ CFU/mL *E. coli*

Current methods:
16-24h to determine
MIC

MCF:
Time for
interpretation
of AST or MIC
results was
established to
correspond to
 $2 \times t_c$ (12h).



500 μ m



5 mm

BMD		Ampicillin conc. (μ g/mL)
1	2	PC
3	4	0.5
5	6	1
7	8	2
9	10	4
11	12	8
13	14	16
15	16	32
17	18	64
19	20	128

Reference mean time to reach the cut-off signal of 0.14 in non-antibiotic microcapillaries as t_c , when the growth rate in antibiotic-containing microcapillaries dropped by 50% (comparison with the non-antibiotic capillary);

Research question and objectives

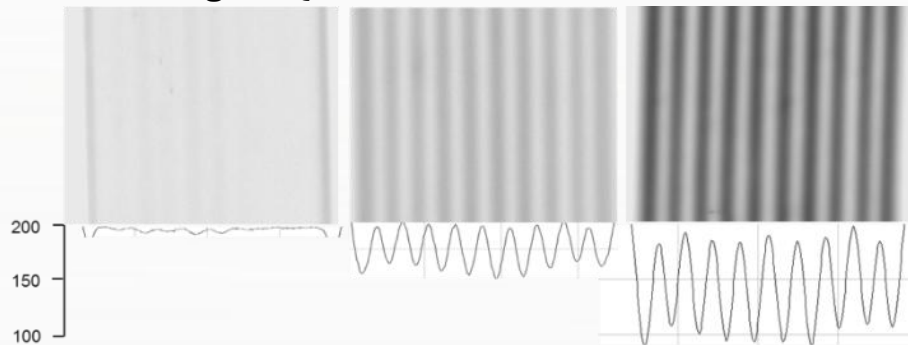
Can machine learning models use **early GVR time-series data** to predict which capillaries in an antibiotic concentration will remain inhibited at the endpoint, allowing early estimation of the MIC (minimum inhibitory concentration)?

$$\text{GVR} = (\text{AbG}(\text{sample}) - \text{AbG}(\text{reference})) / \text{AbG}(\text{reference})$$

AbG (sample): Colour intensity with bacteria and antibiotics.

AbG (reference): Colour intensity of the medium alone.

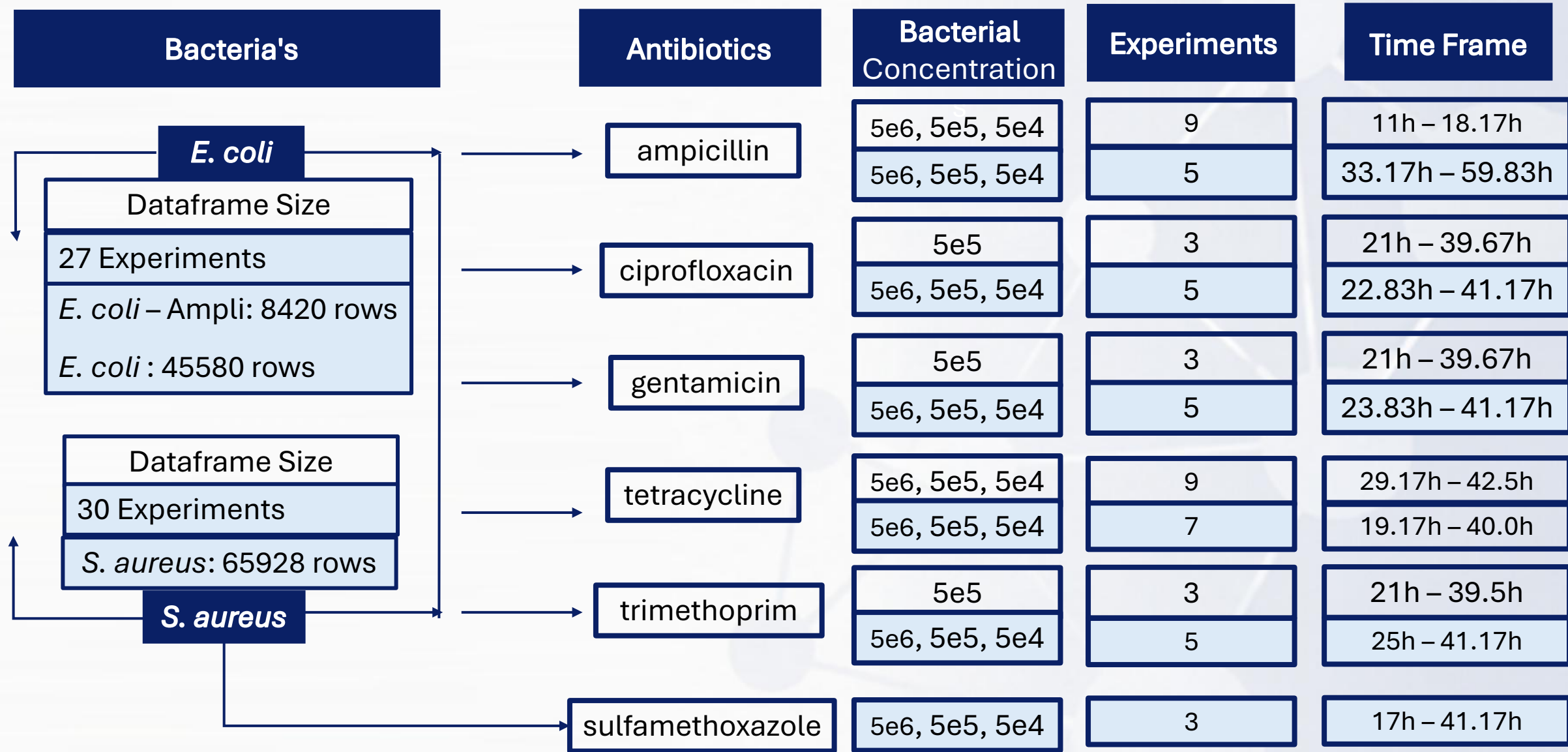
Signal Quantitation



Objectives:

- ✓ Organise data from experiments containing all the GVR related to several combinations of bacteria type, bacteria concentration, time, antibiotic type, and antibiotic concentration in Python.
- ✓ Test different machine learning models using Python software;
- ✓ Design features for the models;
- ✓ Compare different models;
- ✓ Evaluate the models and compare them with experimental data.

Data organisation



Machine learning Features

Experience					Tube id		Antibiotic		Label
Concentration	Time		GVR		Inical Std		Current Std		
Tube Std	Slope		Delta		AUC		Derivative		
Derivative Per Time	Delta Per Time		AUC Per Time		GVR Max Tube		GVR Min Tube		
GVR Min Experience	GVR Max Experience		GVR Average		Growth Score Exp		Growth Score		

Machine learning model training

Data Preparation	Identifiers and antibiotic concentration removed from feature set
Model Selection	Evaluate models - accuracy, ROC-AUC, F1, precision, recall, confusion matrix, cross validation
Temporal Analysis	Best model tested per time point to identify earliest reliable prediction
Feature Refinement	Iterative feature engineering and model optimisation
Validation Strategy	Pipeline developed on <i>E. coli</i> - ampicillin subset, then scaled to full <i>E. coli</i> dataset

Model Comparison

Cross-validation ROC-AUC:
Mean: 0.9443

Random Forest (no concentration) – Test 1

Accuracy : 0.7525252525252525

ROC-AUC : 0.8833568003264973

	precision	recall	f1-score	support
0	0.68	0.96	0.79	495
1	0.93	0.55	0.69	495
accuracy			0.75	990
macro avg	0.80	0.75	0.74	990
weighted avg	0.80	0.75	0.74	990

Confusion Matrix (TN FP / FN TP):

```
[[473 22]
 [223 272]]
```

E. coli Chosen Model

	Accuracy	ROC-AUC
Logistic Regression	0.7505	0.8439
Random Forest	0.7697	0.8718
Support Vector Machine	0.7455	0.7353
K-Nearest Neighbours	0.7293	0.7802
Naive Bayes	0.7535	0.8057

E. coli - Ampicillin Models

	Accuracy	ROC-AUC
Logistic Regression	0.7394	0.8620
Random Forest	0.7525	0.8834
Support Vector Machine	0.7586	0.8391
K-Nearest Neighbours	0.7333	0.8268
Naive Bayes	0.7434	0.8497

E. coli Models

Random Forest (no concentration) – Test 1

Accuracy : 0.7696969696969697

ROC-AUC : 0.8718110396898275

	precision	recall	f1-score	support
0	0.79	0.74	0.76	495
1	0.75	0.80	0.78	495
accuracy			0.77	990
macro avg	0.77	0.77	0.77	990
weighted avg	0.77	0.77	0.77	990

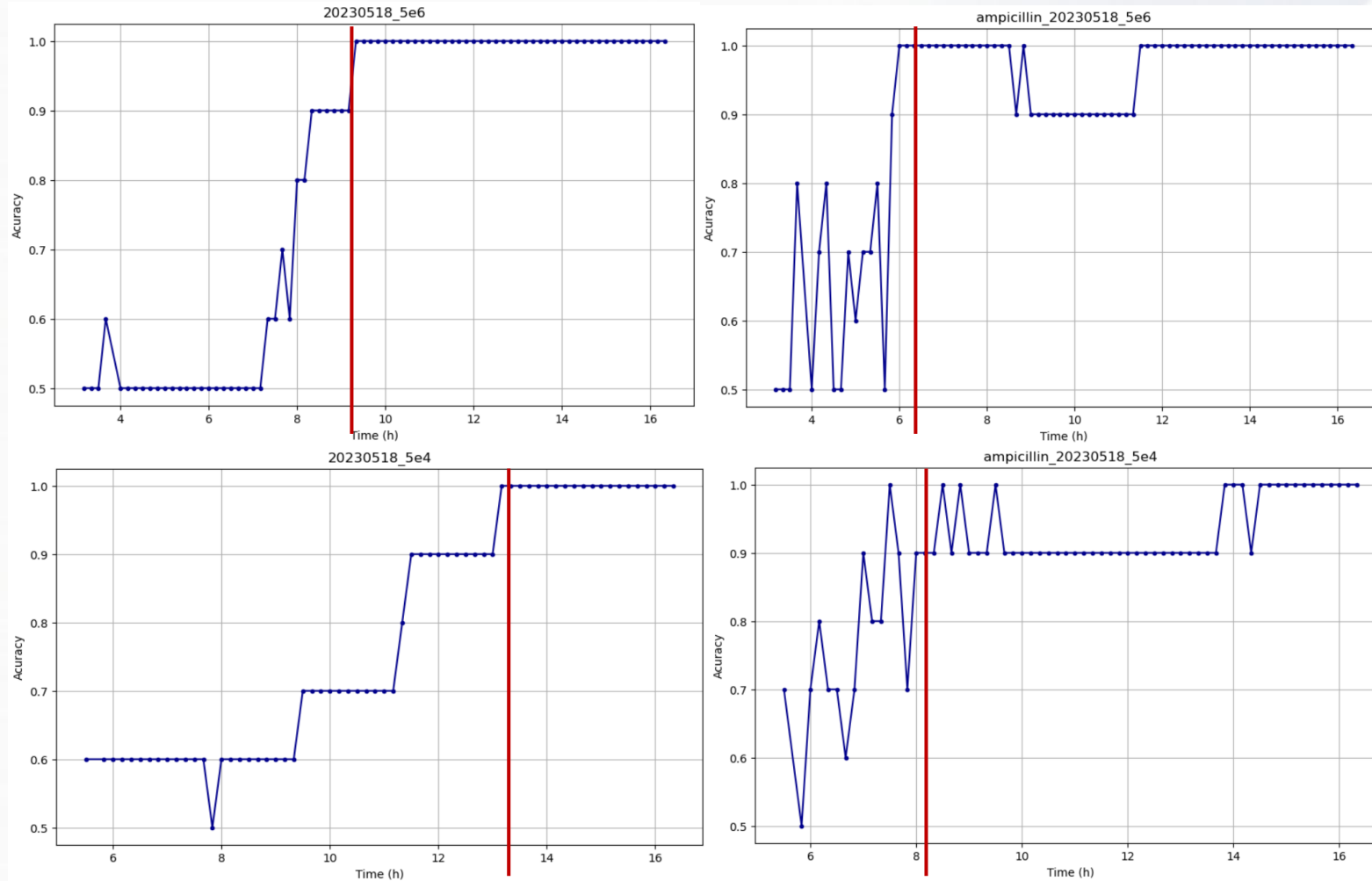
Confusion Matrix (TN FP / FN TP):

```
[[365 130]
 [ 98 397]]
```

Cross-validation ROC-AUC:
Mean: 0.9191

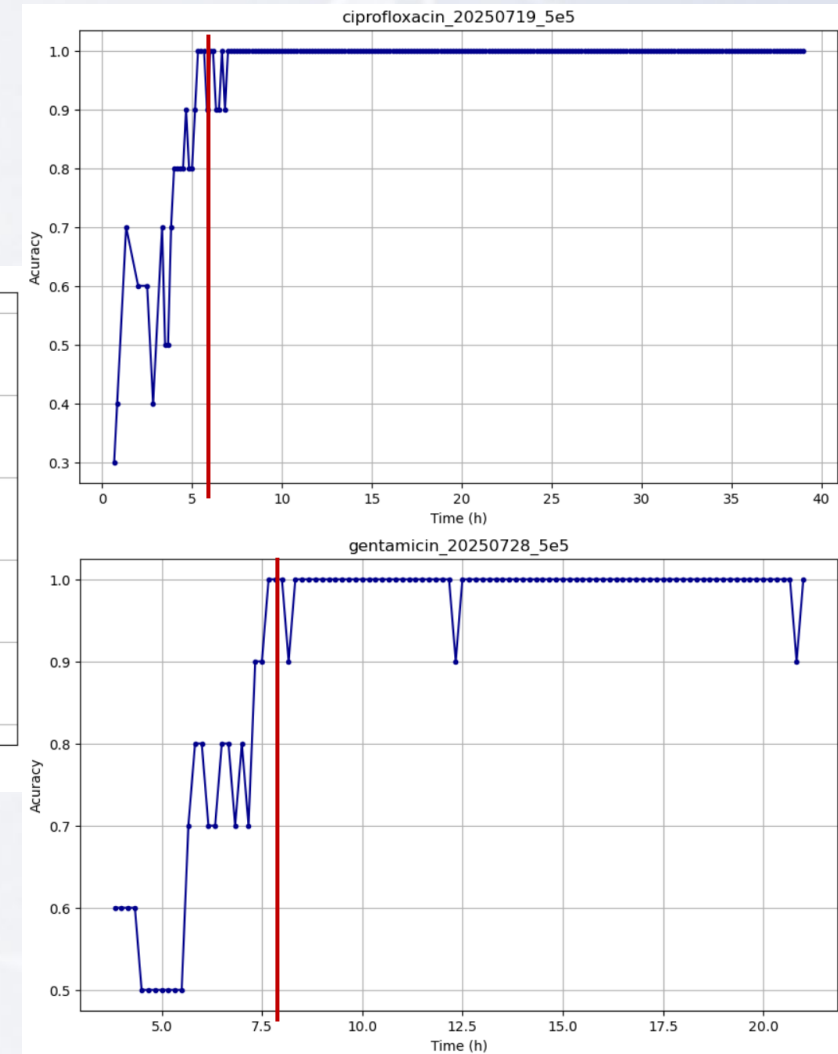
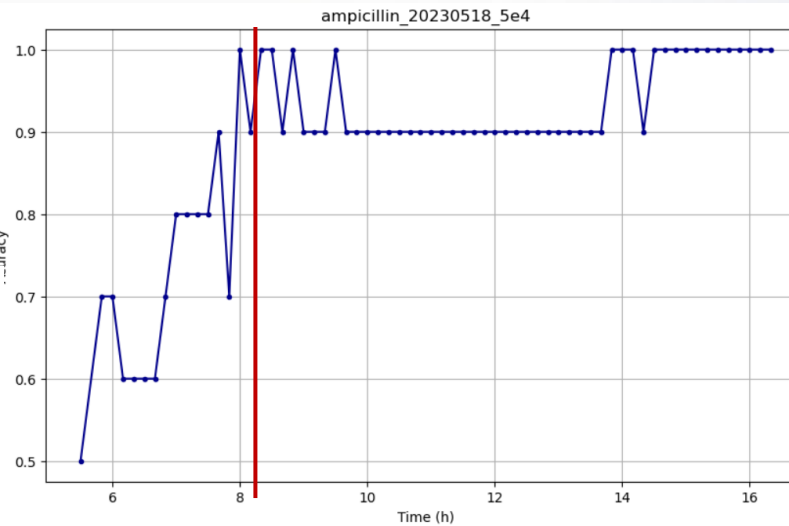
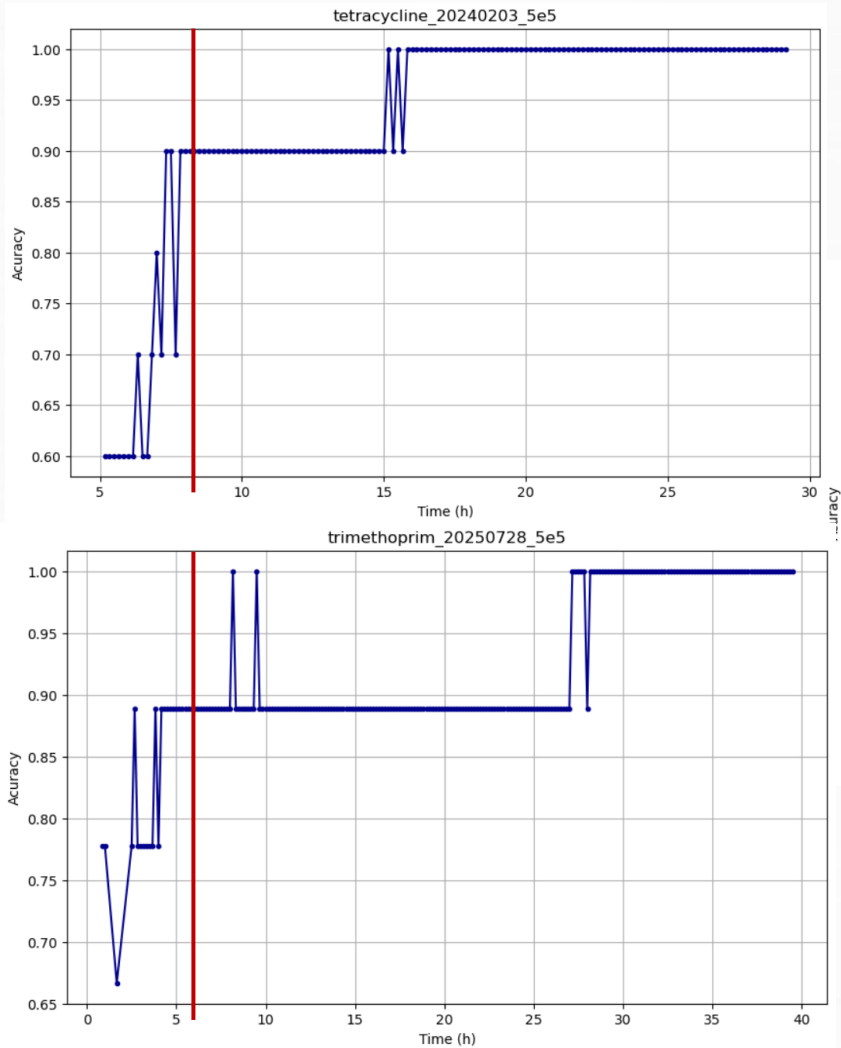
E. coli - Ampicillin Chosen Model

Ampicillin Testing



Accuracy over time: *E. coli* Ampicillin-only training (left) vs. All antibiotics training (right).

All antibiotics testing



Accuracy over time across the five antibiotics (Ciprofloxacin, Gentamicin, Tetracycline, Trimethoprim, Ampicillin) - Model trained with all *E. coli* data.

Conclusion

Traditional
AST

16h – 24h

MC-AST

12h

ML Model

10h



No AI tools were used in the
preparation of this presentation